

## BEAUCEVILLE, QC, Canada

WMO: 713230

Lat: 46.2050N

Lon: 70.7850W

Elev: 229

StdP: 98.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -26.2      | -23.6 | -30.6                         | 0.2 | -25.7 | -28.0 | 0.3 | -23.1 | 7.0                   | -6.5 | 6.1 | -8.6 | 1.7                   | 170  | 0.537 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.8                   | 28.8            | 20.3 | 27.2 | 19.4 | 25.7 | 18.7 | 22.1                | 26.3 | 21.1 | 24.9 | 20.1 | 23.8 | 2.8                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.7                            | 15.8 | 23.9 | 19.7 | 14.9 | 23.1 | 18.8 | 14.0 | 22.1 | 65.9          | 26.1 | 62.2 | 25.0 | 58.8 | 23.8 | 27.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 6.2               | 5.3  | 4.7 | -31.1                      | 31.6 | 3.5                | 1.4 | -33.6                                              | 32.6 | -35.7      | 33.3 | -37.6      | 34.1 | -40.1      | 35.1 |
|                   |      |     | -31.5                      | 23.7 | 3.5                | 1.1 | -34.0                                              | 24.6 | -36.0      | 25.2 | -38.0      | 25.9 | -40.5      | 26.7 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 4.9    | -10.8 | -9.2  | -3.7  | 3.7   | 11.2  | 16.2  | 18.9  | 17.7  | 13.6  | 6.9   | 0.2   | -6.7 |
|                                                                                  | DBStd       | 11.50  | 7.33  | 6.51  | 6.16  | 4.51  | 4.26  | 3.71  | 2.93  | 3.04  | 4.08  | 4.38  | 5.24  | 6.30 |
|                                                                                  | HDD10.0     | 2779   | 644   | 538   | 426   | 194   | 36    | 2     | 0     | 0     | 11    | 116   | 295   | 517  |
|                                                                                  | HDD18.3     | 5007   | 903   | 772   | 684   | 438   | 225   | 86    | 29    | 48    | 152   | 353   | 543   | 775  |
|                                                                                  | CDD10.0     | 922    | 0     | 0     | 1     | 6     | 73    | 186   | 274   | 240   | 118   | 22    | 2     | 0    |
|                                                                                  | CDD18.3     | 107    | 0     | 0     | 0     | 0     | 3     | 21    | 45    | 30    | 9     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 984    | 0     | 0     | 0     | 5     | 66    | 220   | 377   | 234   | 79    | 2     | 0     | 0    |
|                                                                                  | CDH26.7     | 191    | 0     | 0     | 0     | 1     | 11    | 48    | 78    | 37    | 16    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.1    | 2.0   | 2.2   | 2.4   | 2.5   | 2.3   | 2.0   | 1.8   | 1.7   | 1.8   | 2.0   | 2.1   | 1.9  |
| Precipitation                                                                    | PrecAvg     | 1135   | 81    | 67    | 74    | 82    | 94    | 117   | 120   | 122   | 101   | 105   | 83    | 96   |
|                                                                                  | PrecMax     | 1448   | 150   | 129   | 136   | 173   | 160   | 220   | 196   | 237   | 184   | 209   | 123   | 171  |
|                                                                                  | PrecMin     | 844    | 35    | 25    | 30    | 22    | 39    | 49    | 33    | 23    | 39    | 33    | 26    | 51   |
|                                                                                  | PrecStd     | 135    | 29    | 25    | 25    | 41    | 32    | 40    | 34    | 46    | 39    | 45    | 24    | 30   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 8.8   | 8.3   | 13.7  | 22.7  | 28.2  | 30.4  | 31.1  | 29.7  | 28.8  | 22.7  | 16.5  | 8.6  |
|                                                                                  |             | MCWB   | 7.9   | 5.3   | 8.5   | 13.9  | 17.5  | 20.8  | 22.3  | 20.9  | 21.1  | 15.8  | 13.3  | 7.7  |
|                                                                                  | 2%          | DB     | 4.7   | 4.6   | 9.5   | 17.9  | 25.1  | 28.0  | 28.8  | 27.6  | 25.5  | 19.5  | 12.9  | 5.6  |
|                                                                                  |             | MCWB   | 4.0   | 2.7   | 4.7   | 10.1  | 15.4  | 19.4  | 20.7  | 19.9  | 19.1  | 14.4  | 9.9   | 4.5  |
|                                                                                  | 5%          | DB     | 1.9   | 2.5   | 6.7   | 14.3  | 22.3  | 26.0  | 27.0  | 26.0  | 23.2  | 16.5  | 10.0  | 3.2  |
|                                                                                  |             | MCWB   | 1.0   | 1.0   | 3.4   | 8.2   | 14.5  | 18.3  | 20.0  | 19.3  | 18.0  | 12.7  | 7.8   | 2.2  |
|                                                                                  | 10%         | DB     | -0.2  | 0.7   | 4.6   | 11.4  | 19.7  | 23.8  | 25.4  | 24.3  | 20.9  | 14.0  | 7.5   | 1.5  |
|                                                                                  |             | MCWB   | -0.9  | -0.5  | 2.2   | 6.5   | 13.2  | 17.2  | 18.9  | 18.5  | 16.9  | 11.2  | 5.8   | 0.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 7.7   | 6.0   | 9.7   | 15.3  | 19.8  | 22.7  | 23.8  | 22.8  | 22.1  | 17.6  | 13.9  | 8.0  |
|                                                                                  |             | MCDB   | 8.3   | 7.5   | 12.1  | 21.2  | 24.4  | 28.3  | 28.2  | 27.2  | 26.6  | 20.5  | 15.5  | 8.6  |
|                                                                                  | 2%          | WB     | 3.8   | 2.8   | 6.0   | 11.6  | 17.6  | 21.2  | 22.2  | 21.5  | 20.4  | 15.3  | 10.6  | 4.7  |
|                                                                                  |             | MCDB   | 4.2   | 3.9   | 7.8   | 15.3  | 22.5  | 25.5  | 26.4  | 25.1  | 24.1  | 18.1  | 12.0  | 5.4  |
|                                                                                  | 5%          | WB     | 0.9   | 1.0   | 4.0   | 9.1   | 15.9  | 19.7  | 21.2  | 20.5  | 18.9  | 13.3  | 8.1   | 2.4  |
|                                                                                  |             | MCDB   | 1.5   | 2.1   | 6.1   | 13.0  | 20.0  | 23.8  | 25.1  | 23.9  | 22.2  | 16.2  | 9.5   | 3.2  |
|                                                                                  | 10%         | WB     | -1.2  | -0.6  | 2.4   | 7.0   | 14.0  | 18.4  | 20.2  | 19.5  | 17.5  | 11.4  | 5.8   | 0.6  |
|                                                                                  |             | MCDB   | -0.4  | 0.5   | 4.7   | 10.7  | 18.4  | 22.1  | 23.9  | 23.0  | 20.3  | 13.5  | 7.3   | 1.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.8   | 10.8  | 10.3  | 10.2  | 11.9  | 11.4  | 10.8  | 11.1  | 11.3  | 9.3   | 7.7   | 7.9  |
|                                                                                  |             | MCDBR  | 10.7  | 11.5  | 12.9  | 16.4  | 16.6  | 14.8  | 13.1  | 13.3  | 13.8  | 13.8  | 11.8  | 8.2  |
|                                                                                  |             | MCWBR  | 10.6  | 9.7   | 9.2   | 9.8   | 8.5   | 7.5   | 6.3   | 6.8   | 7.9   | 9.1   | 9.5   | 7.9  |
|                                                                                  | 5% WB       | MCDBR  | 10.5  | 10.5  | 10.8  | 14.3  | 13.2  | 12.4  | 11.0  | 10.7  | 12.0  | 12.2  | 10.8  | 8.3  |
|                                                                                  |             | MCWBR  | 10.3  | 9.4   | 8.5   | 9.6   | 7.9   | 7.1   | 6.1   | 6.3   | 7.5   | 8.8   | 9.4   | 8.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.252  | 0.275 | 0.316 | 0.337 | 0.358 | 0.403 | 0.408 | 0.390 | 0.363 | 0.330 | 0.301 | 0.283 |      |
|                                                                                  | taud        | 2.373  | 2.275 | 2.271 | 2.398 | 2.411 | 2.271 | 2.277 | 2.333 | 2.406 | 2.496 | 2.495 | 2.336 |      |
|                                                                                  | Ebn at Noon | 866    | 904   | 910   | 921   | 908   | 863   | 854   | 856   | 851   | 834   | 798   | 769   |      |
|                                                                                  | Edh at Noon | 71     | 96    | 113   | 110   | 113   | 131   | 128   | 117   | 99    | 79    | 65    | 67    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.41   | 2.30  | 3.53  | 4.42  | 5.00  | 5.51  | 5.44  | 4.76  | 3.79  | 2.20  | 1.29  | 0.95  |      |
|                                                                                  | RadStd      | 0.20   | 0.29  | 0.28  | 0.41  | 0.54  | 0.36  | 0.51  | 0.30  | 0.33  | 0.19  | 0.16  | 0.10  |      |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (28 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page